

Daily Logic Drill #7

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Paper 2 Capstone Synthesis · Implication Chains · Modulo Characterisations · Proof Auditing.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Capstone speed recognition: one question per core logic skill from across the week.

1. What is the negation of the proposition “ $\forall x \in \mathbb{R}, x^2 + 1 > 0 \implies x > 0$ ”?
2. For integers n , is “ n is divisible by 6” necessary, sufficient, both, or neither for “ n is divisible by 2”?
3. Consider the implication chain $P \implies Q$, $\neg R \implies \neg Q$, and $R \implies S$. If P is true, what is the truth value of S ?
4. A student solves $\sqrt{2x+3} = x$ by squaring both sides and obtains candidate solutions $x = 3$ and $x = -1$. Which candidate is extraneous?
5. Find the smallest odd composite positive integer that is not a prime power.
6. For real numbers x , is “ $x^2 > 9$ ” necessary, sufficient, both, or neither for “ $x > 3$ ”?
7. A sequence satisfies $u_1 = 2$ and $u_{n+1} = 3u_n - 2$. What is the closed-form expression for u_n ?
8. In a proof by contradiction of the implication $P \implies (Q \vee R)$, what compound statement is assumed alongside P ?
9. Find the smallest positive integer n such that $2^n + 1$ is composite.
10. If $P(1)$ is true and $P(k) \implies P(2k)$ holds for all $k \geq 1$, how many integers in $\{1, 2, \dots, 20\}$ are proven true?

Section B Manipulation Drills

(≤50 s each)

Each question pairs two techniques from different days.

1. Prove by contrapositive: “If n is not a multiple of 3 and not a multiple of 5, then n is not a multiple of 15.”

2. Prove by contradiction: “There is no positive integer n such that $n^2 + n$ is odd.”
3. Prove by induction: $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$ for all positive integers n .
4. A student wants to prove “if n is not a multiple of 5, then n^2 is not a multiple of 25” by instead showing “if n is a multiple of 5, then n^2 is a multiple of 25.” Is this a valid proof of the original claim? Justify your answer.
5. Give a counterexample showing that “ n is prime” is not sufficient for “ n is odd.”
6. Negate the statement: “For every set A , there exists a set B such that $A \cap B = \emptyset$.” Is the negated statement true or false?
7. Prove by induction that $n^3 + 2n$ is divisible by 3 for all positive integers n .
8. Prove by contradiction that $\sqrt{6}$ is irrational.
9. For a polynomial f , classify “ $f(x) > 0$ for all real x ” as necessary, sufficient, both, or neither for “ f has no real roots.”
10. A student proves $P(n)$ for all $n \geq 1$ by checking $P(1)$ directly, then showing $P(k) \implies P(k+1)$ only for $k \geq 2$ (skipping $k = 1$). Explain the gap this leaves in the proof.

Section C Substitution & Structure(≤ 90 s each)*Tricky synthesis MCQs, combining techniques from across the week.*

1. Let P : “ n is divisible by 4” and Q : “ n is divisible by 2 and n is divisible by 6”. Which of the following is true?
 - A) $P \iff Q$.
 - B) P is sufficient but not necessary for Q .
 - C) P is necessary but not sufficient for Q .
 - D) P and Q are neither necessary nor sufficient for each other.
 - E) Q is impossible to satisfy.
2. A theorem states: “If every element of a finite set S of positive integers is even, then the sum of all elements of S is even.” A student, wanting to prove the converse, argues: “Take $S = \{2, 4\}$; the sum is 6, which is even, and every element is indeed even, confirming the converse.” What is the flaw?
 - A) The theorem itself is false.
 - B) A single supporting example does not prove a universal converse claim.
 - C) $S = \{2, 4\}$ does not actually have an even sum.
 - D) The original theorem’s proof technique (not the converse) is what’s flawed.
 - E) There is no flaw; the converse is proved.

3. Which proof technique is most directly suited to establishing: “There exists a positive integer n such that $n^2 + 3n + 1$ is not prime”?
- A) Induction.
 - B) Direct exhibition of a specific such n (e.g. $n = 6$: $6^2 + 3(6) + 1 = 55 = 5 \times 11$).
 - C) Contradiction.
 - D) Contrapositive.
 - E) Strong induction.
4. A student claims: “ n is a multiple of 9” is necessary and sufficient for “the digit sum of n is a multiple of 9”. They verify this only for $n = 9, 18, 27$ and declare the claim proved. Evaluate this.
- A) The claim and the verification are both fully correct.
 - B) Checking finitely many cases cannot prove a universal claim.
 - C) The claim is false; digit sums have nothing to do with divisibility by 9.
 - D) The claim is true only for necessity, not sufficiency.
 - E) The claim is true only for sufficiency, not necessity.
5. What is the contrapositive of “If n^2 is not divisible by 4, then n is odd”?
- A) If n is odd, then n^2 is not divisible by 4.
 - B) If n is even, then n^2 is divisible by 4.
 - C) If n^2 is divisible by 4, then n is even.
 - D) If n is not odd, then n^2 is not divisible by 4.
 - E) If n^2 is divisible by 4, then n is odd.
6. A student claims $P(n)$: “ $n^2 + 5n + 2$ is even for every positive integer n ” is provable by induction: base case $P(1)$: $1 + 5 + 2 = 8$ is even; inductive step: assume $k^2 + 5k + 2$ is even; then $(k + 1)^2 + 5(k + 1) + 2 = k^2 + 7k + 8 = (k^2 + 5k + 2) + 2k + 6$, which is (even)+(even)=even, so $P(k + 1)$ holds. Is this induction valid, and is the underlying claim true?
- A) The induction is valid, and the claim is true for every positive integer.
 - B) The induction has a flaw in the algebra.
 - C) The claim is false — there exists some n for which $n^2 + 5n + 2$ is odd.
 - D) The base case is wrong.
7. Consider positive integers m, n . Statement P : “For every prime p , if $p \mid mn$ then $p \mid m$ or $p \mid n$.” Statement Q : “There exists a prime p such that $p \mid mn$ but $p \nmid m$ and $p \nmid n$.” Which of the following is true?
- A) P and Q can both be true simultaneously.
 - B) Q is the negation of P , so Q is always false because P is always true.
 - C) P is the negation of Q .

D) Neither P nor Q is ever true.

E) P is false in general.

8. A student claims: “ $x = 1$ ” is necessary and sufficient for “ $x^3 - 1 = 0$ ” (for real x), reasoning: “Sufficient, because substituting $x = 1$ gives 0. Necessary, because $1^3 - 1 = 0$, which confirms $x = 1$ works.” Evaluate the necessity argument specifically.

A) Both parts of the reasoning are fully correct as given.

B) The “necessity” argument is circular.

C) The necessity argument is flawed because $x = -1$ is also a solution.

D) Sufficiency is actually false.

E) There is no correct way to establish necessity here.

Section D Challenge

(SMC / BMO1 difficulty)

Full capstone proofs, drawing on every technique from the week.

- Long John Silverman has buried treasure at an integer point (x, y) (not necessarily positive). He records the two values $x^2 + y$ and $x + y^2$; these two recorded numbers are distinct from each other. Prove that no other integer point $(x', y') \neq (x, y)$ can produce the same pair of recorded values (i.e. satisfy both $x'^2 + y' = x^2 + y$ and $x' + y'^2 = x + y^2$) — so the map pins down the treasure’s location uniquely. (*after BMO1 2010 Q6*)
- Two bags contain m and n balls respectively ($m, n > 0$). Two operations are allowed, applied any number of times in any order: (a) remove the same number of balls from each bag; (b) triple the number of balls in one (chosen) bag. (*after BMO1 2012 Q4*)
Prove that if $m - n$ is odd, it is impossible to ever empty both bags (i.e. reach $(0, 0)$) using only these two operations.
- Prove that there do not exist positive integers $a < b < c$ with $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$ and a, b, c all even.
- Prove, using strong induction, that every positive integer $n \geq 2$ can be written as a product of primes in exactly one way, up to the order of the factors (the uniqueness part of the Fundamental Theorem of Arithmetic) — assuming as given Euclid’s Lemma: if a prime p divides a product ab , then $p \mid a$ or $p \mid b$.
- Consider the statement (*): “For all positive integers m, n : m and n are coprime if and only if there exist integers x, y (not necessarily positive) with $mx + ny = 1$ ” (Bézout’s Identity).
(a) Prove the $\Leftarrow$ direction of (*) directly.
(b) Taking the $\Rightarrow$ direction of (*) as given (it is normally proved via the Euclidean algorithm), use (*) to prove: if p is prime and $p \nmid a$, then there exist integers x, y with $px + ay = 1$; hence deduce Euclid’s Lemma (if $p \mid ab$ then $p \mid a$ or $p \mid b$) as a consequence.

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths